LESSON PLAN

ADVANCED AIRCRAFT TECHNOLOGY (20)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	16
Total No. of Lesson Plans	20	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to EFIS, display units, Symbol Generators and display presentations
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to EFIS, display units, Symbol Generators and display presentations. Visit to P8i Lab for familiarization on EFIS.
5. Specific Objectives	20.6.1.1 Trainee Will Be able to explain EFIS units and display presentations.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Visit to P8i Lab for familiarization on EFIS units and display presentations. Safety: Ensure lab is cleared and instructor supervision is maintained during system interaction.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the EFIS?
(b) Name two types of display units found in modern cockpits.
(c) How does a Symbol Generator differ from a display unit?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition and purpose of EFIS • Components: CDU, PFD, ND • Symbol Generator architecture • Data sources for EFIS • Display presentation formats	L & Ds	PPT, IB	15 min
(b)	• P8i Lab EFIS configuration • Navigation Display symbology • Primary Flight Display symbology • Control Display Unit functions • System integration overview	L & Ds	PPT, PDF	20 min
(c)	• Troubleshooting EFIS failures • Reversionary modes operation • Display format selection • Symbol generator redundancy • Data link integration	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the role of the Symbol Generator in EFIS operation.
(b) Describe the difference between PFD and ND display presentations.
(c) How does the CDU interact with the EFIS display units?

12. Home Assignment and Follow Up	Prepare a brief report comparing EFIS and EICAS/ECAM display architectures for the next lesson (LP 17).
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Introduction to Engine indicating and crew alerting system (EICAS) & Electronic centralized Aircraft Monitoring (ECAM). Practical on EICAS & ECAM.	Refer to the master syllabus reference list	LP 17

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min